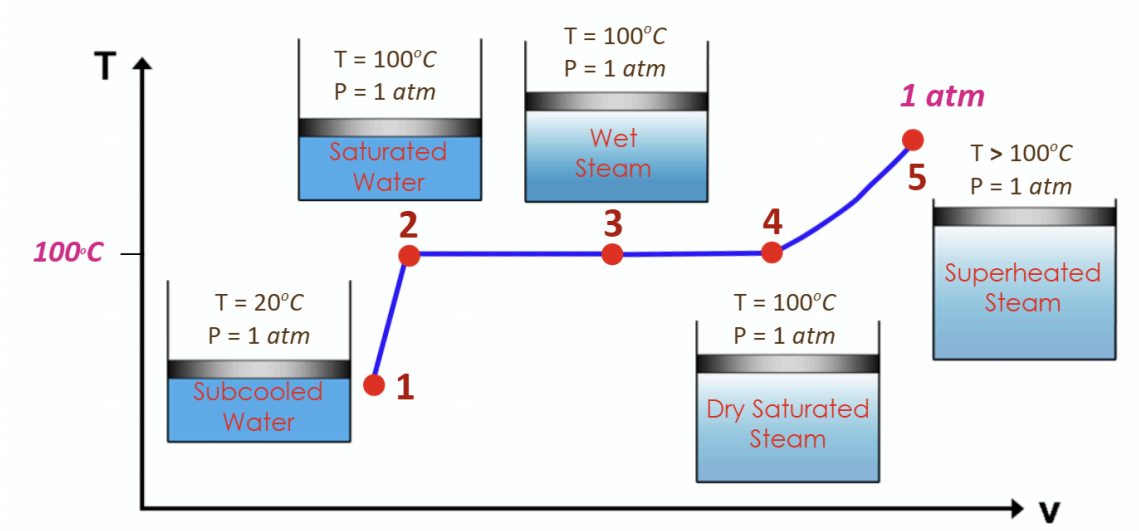
Phases & Properties of Steam

Steam is a complex gas that cannot be accurately represented by ideal gas equations. Hence, we use steam tables to find values we need for steam.

Phase Changes of Steam

Consider water at 20C at 1atm that is gradually heated while keeping the pressure constant. We can plot the temperature and volume of the system like so:



Subcooled Water

When water is initially at the liquid state, we refer to it as subcooled water.

Saturated Water

Once the water reaches the saturation temperature, or the boiling point, water will first start to boil. The water is still liquid at this state, hence we call it saturated water. This state is also referred to as the saturated liquid state.

Wet Steam

As the water continues to boil, there is both liquid water and steam, forming a liquid vapour mixture called wet steam. As the water is still boiling, the

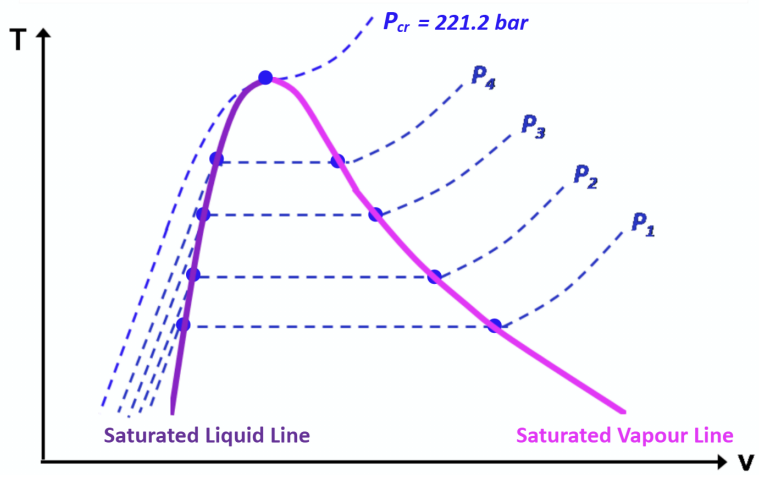
Dry Saturated Steam

When all of the water has vapourised and has finished boiling, there is only dry saturated steam present, with no water. This state is also referred to as the saturated vapour state

Superheated Steam

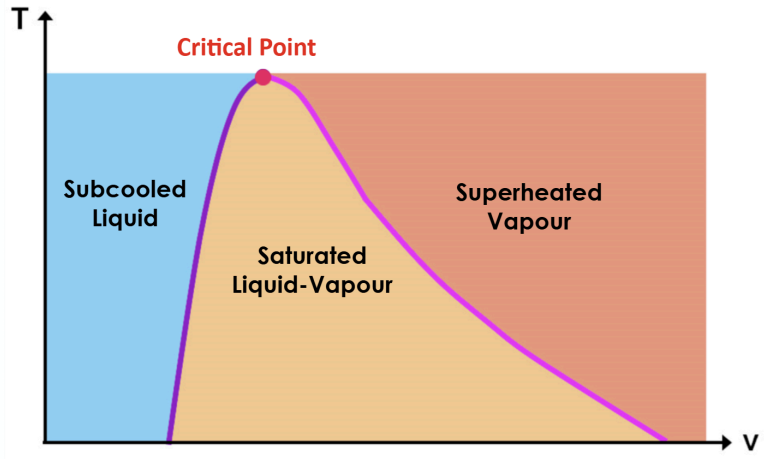
Further heating the steam past its saturated point will result in superheated steam.

If we plot multiple phase changes of steam at different constant pressures on a T-V diagram, we can form the following:



Pcr refers to the critical pressure, where the saturated water and saturated vapour state are the same. This state also forms the critical point.

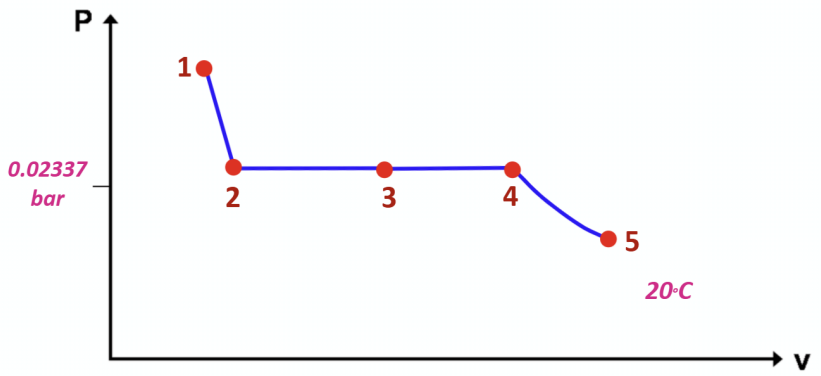
If we join the saturated states, we can form this diagram:



If we keep the temperature constant, while heating it and changing the pressure of the container, we can obtain the phase change of water under

constant temperature

In this case, temperature is kept at 20C

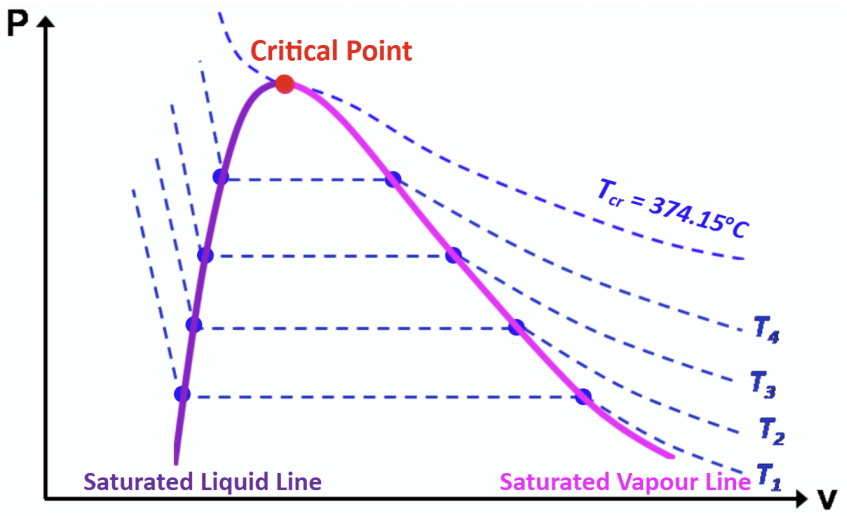


The states are in the same order, however the few things to note is:

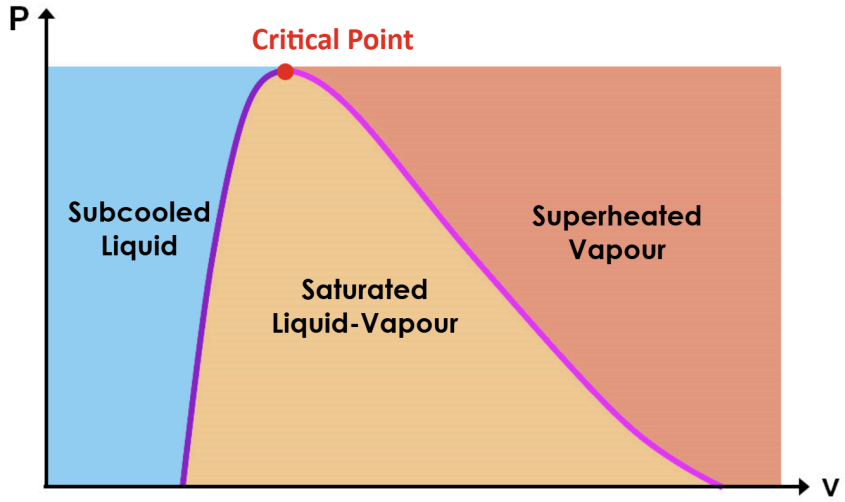
as the water is heated, the pressure is reduced to maintain the same temperature.

When the water is heated to saturated water, it reaches a specific temperature called the saturation pressure. For 20C, the saturation pressure is 0.02337 bar.

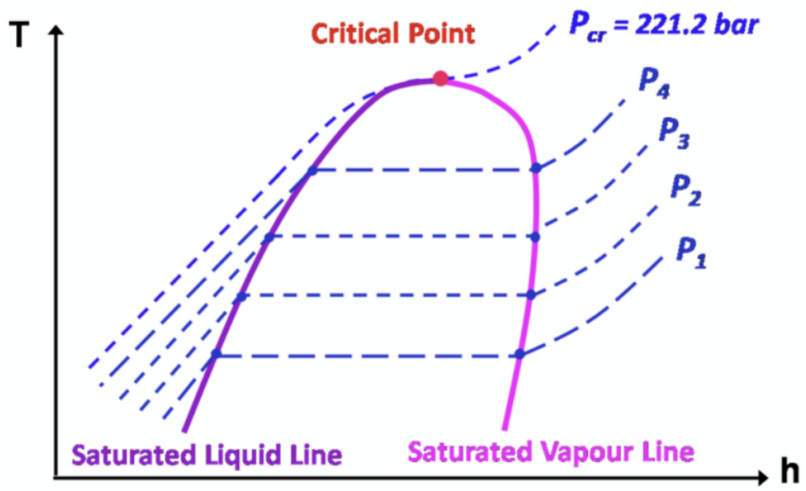
Again, we can plot multiple phase changes at different constant temperatures like so:



And by joining the saturated liquid and vapour states, we can form the P-V diagram:



There are other ways to for diagrams relating the states to different quantities, like the T-h diagram, where pressure is kept constant:



Saturation Temperature & Pressure

As said earlier, the temperature water starts to boil at a specific pressure is the saturation temperature, T_s.

The pressure water starts to boil at a specific temperature is the saturation pressure, P_s.

For example, at 1atm, Ts = 100C, and vice versa.

Generally, Ts is directly proportional to Ps, as a higher pressure requires more energy and heat to get the water molecules to overcome the higher pressure to boil away, and vice versa.

Dryness Fraction

As water boils from saturated water to dry steam, it will reach wet steam. As it boils, there will be more water that boils into dry steam, hence there will be less saturated water in the wet steam as it boils.

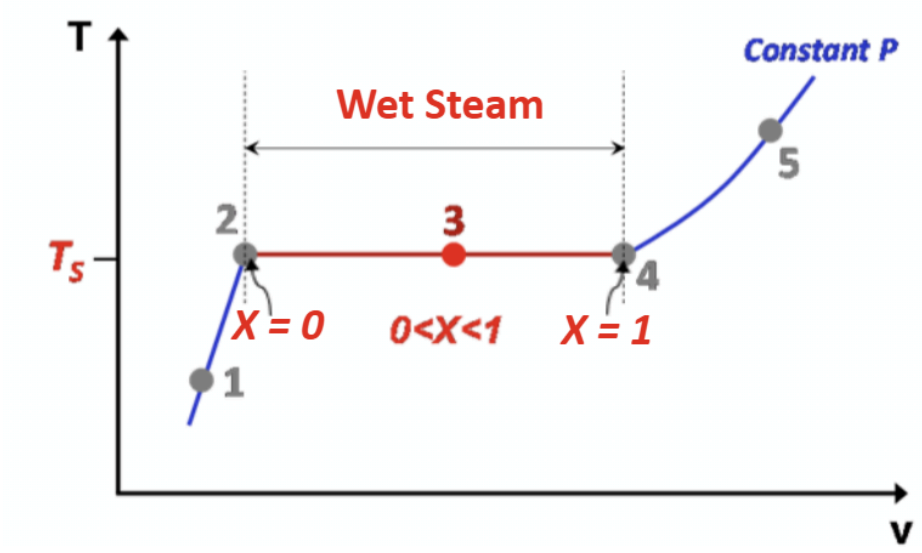
We can quantify this with the dryness fraction, found as:

$$x = \frac{\text{Mass of dry steam}}{\text{Total mass of wet steam}}$$

Note that wet steam is not the amount of water in the steam mixture, but the mixture of dry steam and water (which is wet steam), and that x is a dimensionless quantity.

Looking at the different phases during boiling again, we know that x = 0 at saturated water, as there will be no dry steam at all, while dry steam will have x = 1 as the whole mixture is dry steam.

Hence, wet steam, which we know is between the two states, will have x between 0 to 1.



Degree of Superheat

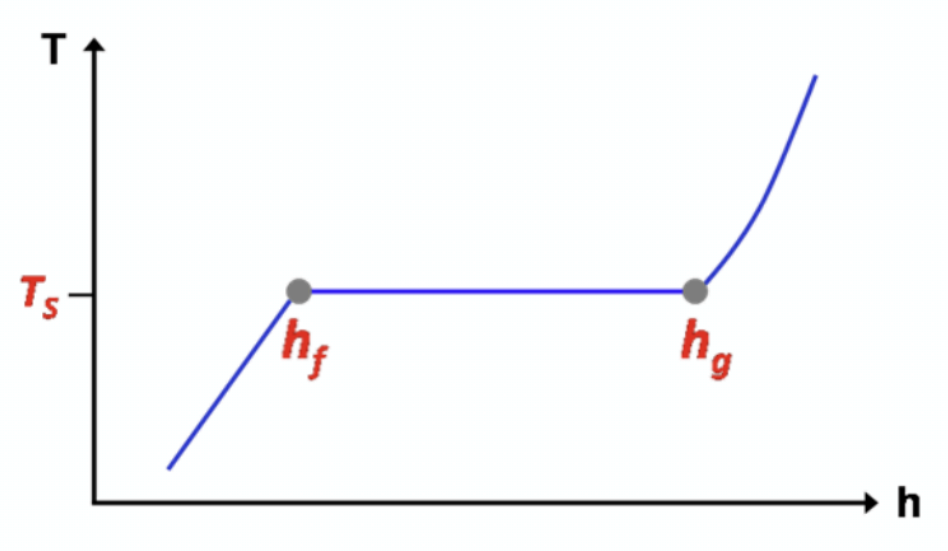
Superheated steam is when dry steam is heated past the saturation temperature. To quantify how 'superheated' the steam is, we can use the degree of superheat, defined as

$$\text{Degree of Superheat} = T - T_s$$

The degree of superheat unit is in degrees Celsius

Specific Enthalpy

For steam, there are 2 important enthalpy values to take note of:



hf is the specific enthalpy of saturated liquid, defined as the energy taken to heat 1kg of liquid from 0C to the saturation temperature to become a saturated liquid.

hg is the specific enthalpy of saturated vapour, defined as the energy needed to heat 1kg of liquid from 0C to the saturated vapour state.

If we take the difference of the two, we get

$$h_{fg} = h_g - h_f$$

which is the specific enthalpy of evaporation. From its definition, we know that its the energy needed to boil the liquid from a saturated liquid to a saturated vapour. This value is also known as the latent heat of vaporisation.